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Under the Guidence of
Amit Katiyar

TAB THIS

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1. If $y = \tan^{-1}\left\{\frac{1+x}{1-x}\right\}$, then $\frac{dy}{dx}$ is equal to .....
(a) $\frac{2}{1+x^2}$ (b) $\frac{1}{1+2x^2}$
(c) $\frac{1-x^2}{1+x^2}$ (d) $\frac{1}{1+x^2}$

2. If $y = \log(\tan x)$, then $\frac{dy}{dx}$ is equal to .....
(a) $2\operatorname{cosec} 2x$ (b) $2\sec 2x$
(c) $2\sin 2x$ (d) $2\cos 2x$

3. If $y = \cos^{-1} x$ and $z = \sin^{-1}\sqrt{1-x^2}$ then $\frac{dy}{dz}$ is equal to ......
(a) $\frac{1}{1-x^2}$ (b) 1
(c) $\frac{1}{1+x^2}$ (d) $\frac{x}{1-x^2}$

4. If $y = e^{2x}$, then $\frac{d^2y}{dx^2}\cdot\frac{d^2x}{dy^2}$ is equal to
(a) $-2e^{x}$ (b) $-2e^{2x}$
(c) $-2e^{-2x}$ (d) $-2e^{-x}$

5. If $\sqrt{x+y} + \sqrt{y-x} = \sqrt{2}$, then $\frac{d^2y}{dx^2}$ is equal to ......
(a) 1 (b) 2
(c) ½ (d) -2

6. $\lim\limits_{x\to 0} \frac{1-\cos x}{x^2}$ is equal to
(a) 0 (b) $\frac{1}{2}$
(c) $\frac{1}{4}$ (d) 1

7. $\lim\limits_{x\to\infty} \left(x - \sqrt{x^2+x}\right)$ is equal to
(a) $\frac{1}{2}$ (b) 1
(c) -1 (d) $-\frac{1}{2}$

8. $\int \frac{dx}{x\log x \log(\log x)}$ is equal to
(a) $\log x$ (b) $\log(\log x)$
(c) $\log(\log(\log x))$ (d) $(\log(\log x))^2$

9. $\int x^x(1+\log x)dx$ is equal to
(a) x^x (b) $x^x \log x$
(c) $\frac{x^x}{\log x}$ (d) $\frac{x^x}{a+x}$

10. $\int_0^1 \frac{x}{(1-x)^{3/4}}dx$ is equal to ......
(a) $12/5$ (b) $-12/5$
(c) $16/5$ (d) $-16/5$

11. Let A and B are two disjoint subsets of a universal set E. The $(A \cup B) \cap B'$ is equal to
(a) E (b) ϕ
(c) A (d) B

12. $(A - B) - A$ is equal to .....
(a) ϕ (b) A
(c) B (d) $A \cap B$

13. Let 10 is the cardinality of set A. The number of bijective mapping from set A to itself is
(a) 10 (b) 55
(c) 100 (d) 3628800

14. Let n be a positive decimal integer. The number of digits in n is equal to ...
(a) $\lceil \log_{10} n \rceil + 1$ (b) $\lfloor \log_{10} n \rfloor + 1$
(c) $\lceil \log_{10} n \rceil$ (d) $\lfloor \log_{10} n \rfloor$

15. Let cardinality of the set A and B are 2 and 5 respectively. The number of relations from A^S to B is ... .
(a) 1024 (b) 1000
(c) 1010 (d) 1025

16. Let $f: R \to R, g: R \to R$ be two functions given by $f(x) = 2x - 3$ and $g(x) = x/2$. The $(f(g))^{-1}(x)$ is equal to ....
(a) $\frac{x+3}{2}$ (b) $x + 3$
(c) $2x + 3$ (d) $2x - 4$

17. Let $f: R \to R$ is defined by $f(x) = x^2 + 5$, then value of $f^{-1}(4)$ is equal to .....
(a) +1 (b) -1
(c) ϕ (d) 20

18. If $g: R \to R$ is defined by $g(x) = x^2 - 2$, then value of $g^{-1}(23)$ is equal to .....
(a) ± 5 (b) 25
(c) ± 4 (d) 527

19. Let cardinality of A and B are 3 and 10 respectively. The number of one one functions from A to B is.
(a) 2^{10} (b) 2^2
(c) 101 (d) 720

20. Let $A = \{1,2,3,4\}$ and $B = \{a, b\}$ are two sets. The number of subjective mappings from A to B is ...
(a) 14 (b) 16
(c) 2^8 (d) 8 !

21. Let $z = \sqrt{3} + i$ be a complex number and $\bar{z}$ be its conjugate. The $|\arg z| + |\arg \bar{z}|$ is equal to
(a) $\frac{\pi}{3}$ (b) $\frac{2\pi}{3}$
(c) $\frac{\pi}{6}$ (d) $\frac{\pi}{4}$

22. The $\frac{(\sqrt{3}+i)^{17}}{(1-i)^{50}}$ is equal to ......
(a) $\frac{-1-\sqrt{3}i}{2^9}$ (b) $\frac{1+\sqrt{3}i}{2^9}$
(c) $\frac{-1-\sqrt{3}i}{2^8}$ (d) $\frac{1+\sqrt{3}i}{2^8}$

23. For which of the following value of x, the $\left(\frac{1+i}{1-i}\right)^x = 1$ is .....
(a) 29 (b) 35
(c) 34 (d) 68

24. If ω is a cube root of unity, then the value of $(1-\omega-\omega^2)(1+\omega^3)$ is
(a) 2 (b) 4
(c) ω (d) ω^2

25. Let z be a complex number. Which of the following is a solution of $|z|-z=1+2i$?
(a) $\frac{3}{2}+2i$ (b) $2-\frac{3}{2}i$
(c) $\frac{3}{2}-2i$ (d) $2+\frac{3}{2}i$

26. If $\sin\theta+\operatorname{cosec}\theta=2$, then $\sin^n\theta+\operatorname{cosec}^n\theta$ is equal to ……
(a) 1 (b) 2
(c) 2^n (d) 2^n-1

27. The value of $\sin^6 x+\cos^6 x+3\sin^2 x\cos^2 x$ is equal to ……
(a) 3 (b) 2
(c) 1 (d) 0

28. IF $x=a\cos^2\theta\sin\theta$ and $y=a\sin^2\theta\cos\theta$ then $(x^2+y^2)^3$ is equal to
(a) a^2x^2 (b) $a^2x^2y^2$
(c) $a^2(y^2-x^2)$ (d) $a^2(x^2-y^2)$

29. The minimum value of $3\cos\theta+4\sin\theta+10$ is equal to
(a) 5 (b) 9
(c) 7 (d) 3

30. $\sin 6°\sin 42°\sin 66°\sin 78°$ is equal to … …
(a) $\frac{1}{32}$ (b) $\frac{1}{16}$ (c) $\frac{1}{8}$ (d) $\frac{1}{4}$

31. If 20^{th} term of an AP is 30 and its 30^{th} term is 20 , then the 10^{th} term is …..
(a) 40 (b) 10
(c) 20 (d) 30

32. Let sum of n terms of an AP is $2n(n-1)$, then the sum of their squares is ……
(a) $\frac{8n(n-1)(2n-1)}{3}$ (b) $\frac{8n(n-1)(2n-1)}{6}$
(c) $\frac{n(n+1)(2n+1)}{6}$ (d) $\frac{8n(n+1)(2n+1)}{3}$

33. For what value of x, the $\log_2(5.2^x+1), \log_4(2^{1-x}+1)$ and 1 are in AP ?
(a) $\log_2 5$ (b) $\log_5 2$
(c) $1+\log_2 5$ (d) $1-\log_2 5$

34. If the ratio of sum of m terms and n terms of an AP be $m^2:n^2$, then the ratio of the m^{th} and n^{th} term will be
(a) $m:n$ (b) $2m-1:2n-1$
(c) $m+n:n+1$ (d) $n:m$

35. The value of $9^{1/3}\times 9^{1/9}\times 9^{1/27}\times\ldots\infty$ is …..
(a) 3 (b) 9
(c) 1 (d) ∞

36. If α and β are the roots of equation $x^2+px+p^2+q=0$, then the value $\alpha^2+\alpha\beta+\beta^2$ …
(a) p (b) $-p$
(c) q (d) $-q$

37. If the roots of $x^2-bx+c=0$ are two consecutive numbers, then b^2-4c is equal to
(a) 1 (b) 2
(c) 3 (d) 4

38. The number of the real roots of the equation $(x-1)^2+(x-2)^2+(x-3)^2=0$ is …
(a) 0 (b) 1
(c) 2 (d) 3

39. If the roots of the equation $(b-c)x^2+(c-a)x+(a-b)=0$ be equals, then a,b,c are in …
(a) HP (b) GP
(c) AP (d) None of these

40. If the equations $x^2+2x+3\lambda=0$ and $2x^2+3x+5\lambda=0$ have a non - zero common root, then λ is equal to ……
(a) 1 (b) -1
(c) 2 (d) -2

41. If ${}^nP_r={}^nP_{r+1}$ and ${}^nC_r={}^nC_{r-1}$, then (n,r) is …
(a) (2,3) (b) (3,2)
(c) (4,3) (d) (3,4)

42. The number of arrangements of the letters of the word BANANA in which the two letters N do not appear adjacently is …..
(a) 40 (b) 60
(c) 80 (d) 100

43. The sum $(n+1)$ terms of the series $\frac{C_0}{2}-\frac{C_1}{3}+\frac{C_2}{4}-\frac{C_3}{5}+\cdots$ is
(a) $\frac{1}{n+1}$ (b) $\frac{1}{n+2}$
(c) $\frac{1}{n(n+1)}$ (d) $\frac{1}{(n+1)(n+2)}$

44. If ω is a cube root of unity, then $\begin{vmatrix}1 & \omega & \omega^2\\ 1 & \omega^2 & 1\\ \omega & 1 & \omega^2\end{vmatrix}$ is equal to …
(a) ω (b) ω^2
(c) 0 (d) -3

45. If $A=\begin{bmatrix}x & 2\\ 2 & x\end{bmatrix}$ and $|A^2|=0$, then x is equal to …
(a) ±2 (b) ±3
(c) 1 (d) 4

46. Let $\vec{A}=i-j+k, \vec{C}=-i-j$ be two vectors. Which of the following is the vector $\vec{B}$ such that

$\vec{A} \times \vec{B} = \vec{C}$ and $\vec{A} \cdot \vec{B} = 1$?

(a) i (b) k
(c) $-j$ (d) $i+j$

47. A point P on y-axis is equidistance from the points $A(-5,4)$ and $B=(3,-2)$. Its coordinate is

(a) $\left(0,\frac{3}{4}\right)$ (b) $\left(0,\frac{4}{3}\right)$
(c) $\left(0,\frac{3}{7}\right)$ (d) $\left(0,\frac{7}{3}\right)$

48. The area of the triangle with vertices $A(a,b+c), B(b,c+a), C(c,a+b)$ is equal to ...

(a) 0 (b) $ab+bc+ca$
(c) $a+b+c$ (d) $a+b-c$

49. Two dices are thrown simultaneously. The probability of obtaining a total score of 5 is ...

(a) $\frac{1}{12}$ (b) $\frac{1}{36}$ (c) $\frac{1}{9}$ (d) $\frac{1}{8}$

50. Three of the six vertices of a regular hexagon are chosen at random. The probability that triangle formed with these chosen vertices is equilateral, equal to

(a) $\frac{1}{2}$ (b) $\frac{1}{10}$
(c) $\frac{1}{5}$ (d) $\frac{1}{20}$

51. Minimum number of two-input NAND gates used to perform the function of two-input OR gate is ...

(a) One (b) Two
(c) Three (d) Four

52. The time required for an electronic circuit to change its state is called ......

(a) propagation Time (b) Rise Time
(c) Decay Time (d) Changing Time

53. Which of the following is not equivalent to x ?

(a) $x \cdot x$ (b) $x+x$
(c) $x.1$ (d) $x+1$

54. Which of the following is a sequential circuit?

(a) Adder (b) Decoder
(c) Multiplexer (d) Flip Flop

55. Which of the following will be the number of output lines in a combinational circuit that takes input a two bit number and produce the output cube of it?

(a) 3 (b) 4
(c) 5 (d) 6

56. Which of the following is a web browser?

(a) Avira (b) TrustPort
(c) Opera (d) None of these

57. Which of the following is an operating system?

(a) Baidu (b) Symbian
(c) AVG (d) None of these

58. Which of the following is antivirus software?

(a) Symbian (b) Norton
(c) AVG (d) None of these

59. Which of the following is a web search engine?

(a) Opera (b) Symbian
(c) AVG (d) None of these

60. Which of the following is a social media website?

(a) Instagram (b) Norton
(c) Symbian (d) None of these

61. z/OS is a

(a) PC operating system
(b) Mainframe operating system
(c) Mobile operating system
(d) None of these

62. Which of the following is a mobile operating system?

(a) Palm operating system
(b) AVG
(c) BeOS
(d) None of these

63. Intel 8086 is a bit microprocessor.

(a) 4 (b) 8
(c) 16 (d) 32

64. Which of the following is mainframe computer?

(a) Vtech (b)Rabbit
(c) Dubna (d)IBM System/360

65. Wellwer is a

(a) Operating System (b) Microprocessor
(c) Mobile company (d) None of these

66. If $(500)_{10} = (x)_5$, then is equal to

(a) 400 (b) 4000
(c) 1000 (d) None of these

67. If $(780)_{10} = (1056)_x$, then x is equal to

(a) 7 (b) 5
(c) 8 (d) 9

68. If $(2?1)_7 = (120)_{10}$, then the missing digit is

(a) 1 (b) 2
(c) 3 (d) 4

69. The 2's complement of the binary number $(0110100)_2$ is

(a) 1001100 (b) 1101100
(c) 1111100 (d) 1101011

70. The 2's complement 10110010 represent the negative number in 8 bits system

(a) -50 (b) -78
(c) -77 (d) -51

71. Which of the following term is wrong in the series $2, 5, 8, 12, 14, 17, 20, ?$

(a) 1st (b) 2nd
(c) 3rd (d) 4th

72. Which of the following term is wrong in the series $1, 4, 9, 16, 21, 36, 49$?
(a) 6^{th} (b) 5 th
(c) 4 th (d) 3^{rd}

73. Which of the following term is wrong in the series $1, 3, 6, 11, 15, 21, 28$
(a) 1st (b) 2nd
(c) 3rd (d) 4th

74. Which of the following is the next term of the series: A_1B, B_2D, D_3G, G_4K, ...?
(a) K_5M (b) K_5P
(c) K_5O (d) K_5Q

75. Which of the following is the next term of the series: $C_1Z, D_3Y, E_5X, F_7W, \dots$?
(a) G_8V (b) $G_{10}V$
(c) G_9W (d) None of these

76. Which of the following is the next term of the series: ABZ, BDY, DFX, GHW,... ?
(a) KJV (b) KIV
(c) JJV (d) JIV

77. Which of the following is the next term of the series: CAT, EBS, GCR, IDQ ...?
(a) KFP (b) KEQ
(c) KEP (d) LEP

78. If '234' is coded to ' 11 ', then '123' is coded to ...
(a) 6 (b) 5
(c) 7 (d) 8

79. If ' 123456 ' is coded to ' 615 ', then ' 214652 ' is coded to ...
(a) 816 (b) 2134
(c) 613 (d) 713

80. $234 : 24 : : 235 : ?$
(a) 9 (b) 56
(c) 210 (d) 30

81. $123 : 9 : : 321 : ?$
(a) 5 (b) 9
(c) 8 (d) 6

82. Which of the following is code for CAT in a coding scheme in which JMI is coded as 32 ?
(a) 21 (b) 24
(c) 23 (d) 22

83. Which of the following is code for JMI in a coding scheme in which BAG is coded as 217 ?
(a) 10139 (b) 9128
(c) 10138 (d) 10129

84. If CAT mean 3, HE mean 2, DELHI mean 5, then SAD is
(a) 1 (b) 2
(c) 3 (d) 4

85. If $54 + 43 = 2, 60 + 51 = 10, 70 + 61 = 12$, then $72 + 62 = ?$
(a) 14 (b) 13
(c) 8 (d) 9

86. Which of the following is next number in the series $1, 3, 6, 11, 18, 29 \dots$?
(a)39 (b) 40
(c) 41 (d) None of these

87. Which of the following is next number in the series $1, 8, 27, 64, 125, \dots$?
(a) 216 (b) 215
(c) 210 (d) None of these

88. Which of the following is next number in the series $3, 7, 13, 21, 31, \dots$?
(a) 41 (b) 43
(c) 47 (d)) None of these

89. Which of the following is next number in the series $1, 2, 6, 42, \dots$?
(a) 57 (b) 1805
(c) 1806 (d) None of these

90. Which of the following term is wrong in the series $1, 1, 2, 4, 5, 8, 13$?
(a) 2^{nd} (b) 4^{th}
(c) 5^{th} (d) 3^{rd}

91. There are views on the issue of giving bonus to the employees.
(a) independent (b) divergent
(c) modest (d) adverse

92. Before the of the Europeans in India, India was a free country.
(a) entry (b) emigration
(c) advent (d) immigration

93. Which of the following is correctly spelt English word?
(a) Delineate (b) Deleneat
(c) Dileneate (d) Deleneate

94. Which of the following is correctly spelt English word?
(a) Enemyty (b) Enemity
(c) Enmity (d) Enmety

95. Which of the following word is most nearly the same in meaning as the word AMAZING?
(a) Beautiful (b) Good
(c) Astonishing (d) Famous

96. Which of the following word is most nearly the same in meaning as the word BRAVE?
(a) Courageous (b) Serene
(c) Aloof (d) Sob

97. Which of the following word is most nearly the same in meaning as the word DILIGENT?
(a) Fool (b) Unhappy
(c) Hardworking (d) Cool

98. Which of the following word is most nearly the opposite in meaning as the word ABSTAIN?
(a) Refrain (b) Desist
(c) Hoard (d) Begin

99. Which of the following word is most nearly the opposite in the meaning as the word MITIGATE?
(a) Aggravate (b) Reduce
(c) Weaken (d) Ease

100. Which of the following word is most nearly the opposite in the meaning as the word AMBIGUOUS?
(a) Opaque (b) Clear
(c) Obscure (d) Vague

ANSWER KEY

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
d	a	b	c	a	b	d	c	a	c
11.	12.	13.	14.	15.	16.	17.	18.	19.	20.
c	a	d	b	a	b	c	a	d	a
21.	22.	23.	24.	25.	26.	27.	28.	29.	30.
a	a	d	b	c	b	c	b	a	b
31.	32.	33.	34.	35.	36.	37.	38.	39.	40.
a	a	d	b	a	d	a	a	c	b
41.	42.	43.	44.	45.	46.	47.	48.	49.	50.
b	a	d	c	a	b	d	a	c	b
51.	52.	53.	54.	55.	56.	57.	58.	59.	60.
c	a	d	d	c	c	b	b,c	d	a
61.	62.	63.	64.	65.	66.	67.	68.	69.	70.
b	a	c	d	d	b	d	c	a	b
71.	72.	73.	74.	75.	76.	77.	78.	79.	80.
d	b	d	b	d	a	c	b	d	d
81.	82.	83.	84.	85.	86.	87.	88.	89.	90.
a	b	a	c	d	d	a	b	c	b
91.	92.	93.	94.	95.	96.	97.	98.	99.	100.
b	c	a	c	c	a	c	d	a	b

ANSWER:

1. $y = \tan^{-1}\left\{\frac{1+x}{1-x}\right\}$

Let x = tan θ

$\because \theta = \tan^{-1}(x)$

$y = \tan^{-1}\left(\frac{1+\tan\theta}{1-\tan\theta}\right)$

$y = \tan^{-1}\left\{\tan\left(\frac{\pi}{4}+\theta\right)\right\}$

$y = \frac{\pi}{4}+\theta$

Putting value of θ

$Y = \frac{\pi}{4} + \tan^{-1}x$

$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{\pi}{4}\right) + \frac{d}{dx}(\tan^{-1}x)$

$\frac{dy}{dx} = 0 + \frac{1}{1+x^2}$

$\frac{dy}{dx} = \frac{1}{1+x^2}$

Ans. (d) $\frac{1}{1+x^2}$

2. y = log (tan x)

$Y = \frac{1}{\tan x}\sec^2 x$

$\frac{dy}{dx} = \frac{\sec^2 x}{\tan x}$

= 2 cosec (2x)

Ans. (a)

3. $y = \cos^{-1}x$ and $z = \sin^{-1}\sqrt{1-x^2}$, $\frac{dy}{dz} = ?$

$\frac{dy}{dz} = -\frac{1}{\sqrt{1-x^2}}$, $x = \cos\theta \Rightarrow \theta = \cos^{-1}x$

$z = \sin^{-1}\sqrt{1-\cos^2\theta}$

$z = \sin^{-1}\sin\theta$

$z = \theta \quad \Rightarrow \quad z = \cos^{-1}x$

$\frac{dz}{dx} = \frac{-1}{\sqrt{1-x^2}}$

$\frac{dy}{dx} \times \frac{dx}{dz} = \frac{-1/\sqrt{1-x^2}}{-1/\sqrt{1-x^2}}$

$\frac{dy}{dz} = 1$

4. $y = e^{2x}$ ------------(1)

$\frac{dy}{dx} = e^{2x}$

$\frac{d^2y}{dx^2} = 20^{2x}.2$

$= 4e^{2x}$

Diff. eq. (1) w.r.t y

$\Rightarrow 1 = 2e^{2x}\frac{dx}{dy}$

$\frac{dx}{dy} = \frac{1}{2e^{2x}} = \frac{1}{2e^{2x}} = \frac{1}{2y} \quad (\because y = e^{2x})$

$\frac{d^2x}{dy^2} = \frac{1}{2}(-1)y^{-2}$

$= \frac{-1}{2}(e^{2x})^{-2} \quad (\because y = e^2)$

$= \frac{-1}{2e^{4x}}$

So $\left(\frac{d^2y}{dx^2}\right)\left(\frac{d^2x}{dy^2}\right) = 4e^{2x} \times \left(\frac{-1}{2e^{yx}}\right)$

$= -2e^{-2x}$

5. $\sqrt{x+y} + \sqrt{y-x} = \sqrt{2}$

$\sqrt{x+y} = \sqrt{2} - \sqrt{y-x}$

$x + y = 2 + (y - x) - 2\sqrt{2}\sqrt{y-x}$

$2x - 2 = -2\sqrt{2}\sqrt{y-x}$

$x - 1 = -\sqrt{2}\sqrt{y-x}$

$(x-1)^2 = 2(y-x)$

2 (x–1) = 2[dy\dx –1]

x-1 + 1 = dy /dx

dy/ dx = x

$d^2y/dx^2 = 1$ Answer.

6. $\lim\limits_{n\to 0} \frac{1-\cos x}{x^2}$

$\lim\limits_{n\to 0} \frac{2\sin\frac{x}{2}}{\frac{x}{2}}$.

$2\lim\limits_{n\to 0} \frac{\sin\frac{x}{2}}{\frac{x}{2}} . \frac{\sin\frac{x}{2}}{\frac{x}{2}}$ x

$x\,\frac{1}{4}\lim\limits_{n\to 0} \frac{\sin\frac{x}{2}}{\frac{x}{2}} = 1$

$2 \times \frac{1}{4}$

= ½ Ans. (b)

7. $\lim\limits_{n\to\infty} \left(x - \sqrt{x^2+x}\right)$

$\lim\limits_{n\to\infty} \frac{(x-\sqrt{x^2+x})(x+\sqrt{x^2+x})}{(x+\sqrt{x^2+x})}$

$\lim\limits_{n\to\infty} \frac{x^2-x^2+x}{(x+\sqrt{x^2+x})} \Rightarrow \frac{+x}{x\left(1+\sqrt{1+\frac{1}{x}}\right)}$

limit apply

$+\frac{1}{2}$

8. $\int \frac{dx}{(\log x).x\log(\log x)}$

Put t = log (log x)

$= \frac{dt}{dx} = \frac{1}{x\,Log\,x}$

= x log xdt = dx

$\int \frac{dt}{t} = log|t| + C$

log | log (log x)| +C

Ans. (c)

9. $\int x^x(1+\log x)dx$

Let $x^x = t$

$d(x^x) = dt$

$d(e^{x\log x}) = dt$

$e^{x\log x}$ (log x +1) dx = dt

x^x (1+ log x) dx = dt

So $\int x^x(1+\log x)dx$

$\int dt = t + c$

$= x^x + C$

Ans. (a)

10. $\int_0^1 \frac{x}{(1-x)^{3/4}}\,dx$

$\int_0^1 \frac{1-x}{[1-(1-x)]^{3/4}}\,dx$

$\int_0^1 \frac{1-x}{x^{-3/4}}\,dx - \int_0^1 x^{1/4}\,dx$

$= \left[\frac{x^{1/4}}{1/4}\right]_0^1 - \left[\frac{x^{5/4}}{5/4}\right]_0^1$

$= 4 - \frac{4}{5} = \frac{16}{5}$

Ans. (c) $\frac{16}{5}$

11. since A & B are disjoint

$(A \cup B) \cap B' = (A \cap B') \cup (B \cap B')$

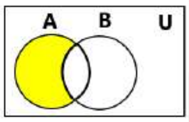
$= A \cup \phi$

= A

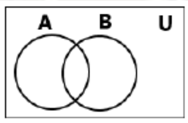
Ans. (c) A

12.

A-B=



(A–B)–A =



Ans. $\emptyset$

13. let a = $\{a_1, a_2, a_3 \ldots\ldots a_{10}\}$

a_1 has 10 choices

a_2 has 9 choices

a_{10} has 2 choices

no. of bichoices

= 10 x 9 x 8 x 7 x 6 x 5 x 4 x 3 x 2 x 1

= 101 = 3628800

Ans. (d)

14. Let n be a +ve integer

Then n is a k digit no.

$= 10^{K-1} \leq n < 10^k$

$K - 1 \leq Log_{10}(n) < k$

$K - 1 = [\log_{10}(n)]$

$K = [\log_{10}(n)] + 1$

Ans. (a)

Q. Corrections ⇒ (b) $[\log_{10}n^2]-1$
(d) $-[\log_{10}n]$

15. n(A) = 2
n (B) = 5
no. of relations = 2^{2x5}
$= 2^{10}$
$= 1024$
Ans. (a)

16. f : R - R g: R - R
f(x) = 2x – 3, g(x) = x/2 $f(g(x)^{-1} = ?$
f(g(x)) = x – 3
y = x – 3
x = y + 3
y = x + 3 ⇒ $f(g(x))^{-1} = x + 3$

17. f : R - R
f(x) = $x^2 + 5$ f' (4) = ?
y = $x^2 + 5$
y –5 = x^2 ⇒ x= $\sqrt{y-5}$ = $f^{-1}(x) = \sqrt{x-5}$
$f^{-1}(4) = \sqrt{4-1} = \sqrt{-1} = \phi$

18. g : R →R g(x) = $x^2 - 2$ g^{-1} (23) = ?
y = $x^2 - 2$ ⇒ $x^2 = y + 2 = 0$ x = $\sqrt{y+2}$
$g^{-1}(x) = \sqrt{x+2}$
$g^{-1}(23) = \sqrt{25}$ ⇒ $g^{-1}(23) = +5$

19. |B| = 10 |A| = 3
No. of injective functions
$\frac{|B|!}{(|B|)!-|A|!}$
$= \frac{10!}{(10-3)!} = \frac{10!}{7!}$
$\frac{10\times9\times8\times71}{7!}$
= 720
Ans. (d)

20. No. of surfection
$= \sum_{r=1}^{n}(-1)^{n-r}\ {}^nC_r\, r^m$
m = 4 n = 2
No. of surfection
$= \sum_{r=1}^{2}(-1)^{2-r}\ {}^2C_r\,(r)^4$
$= (-1)^{2-1}\ {}^2C_1\,(1)^4 + (-1)^{2-2}\ {}^2C_2\,(2)^4$
= –2 + 16
= 14
Ans. (a) 14

21. z = $\sqrt{3} + i$
Arg z = $\tan^{-1}\frac{(imaginary\ part\ of\ z)}{Real\ part\ of\ z}$
$= \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$
$\bar{z}$ is confecgate of z
Is $\sqrt{3} - i$
So arg $\bar{z} = \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right)$
$= \frac{-\pi}{6}$
|arg z| + |arg $\bar{z}$ | = |π/6| + |-π/6|
$= \frac{\pi}{6} + \frac{\pi}{6}$
$= \frac{\pi}{3}$
Ans. (a)

22. $\frac{(\sqrt{3}+i)^{17}}{(1-i)^{50}}$
$\Rightarrow \frac{-1-\sqrt{3}\ i}{2^9}$

23. $\left(\frac{1+i}{1-i}\right)^x = 1$
$\frac{1+i}{1-i}\times\frac{1+i}{1+i} = \frac{1+i^2+2i}{1-i^2}$
$= \frac{1-1+2i}{1+1} = \frac{2i}{2}$
$\frac{1+i}{1-i} = i$
Then
$\left(\frac{1+i}{1-i}\right)^x = (i)^x = 1$
$(i)^x = (i)4$ or $(i)^8$ or $(i)^{12}$ or $(i)^{4th}$
X = 4∩ for ∩ ∈ I
Ans. (d) 68

24. $(1-w-w^2)(1+w^3)$
Cube root of unity
1,w, w^2
$1 + w^3 = 1 + 1 = 2$ (∵ w^3=1)
$1-w-w^2 = 1-(w+w^2)$
1–(–1) = 2
So $(1-w-w^1)(1+w^3)$
= 2.2 = 4
Ans. (b)

25. |Z| – Z = 1+ 2i
$= \sqrt{x^2+y^2} - (x+iy)$
1+2i (z= x + iy)
$= \sqrt{x^2+y^2} - 1 = 1$
Y = –2
Compairing real and imaginary parts
X = 3/2 y = –2
Ans. (c) $\frac{3}{2} - 2i$

26. sin θ + cosec θ = 2, $\sin^n\theta + \mathrm{cosec}^n\theta$ = ?
$(\sin\theta + \mathrm{cosec}\theta)^2 = 4$

$\sin^2\theta + \csc^2\theta + 2\sin\theta\csc\theta = 4$

$\sin^2\theta + \csc^2\theta = 2$

$(\sin\theta + \csc\theta)^3 = 8$

$\sin^3\theta + \csc^3\theta + 3\sin\theta\csc\theta(\sin\theta + \csc\theta) = 8$

$\sin^3\theta + \csc^3\theta + 3 \times 2 = 8$

$\sin^3\theta + \csc^3\theta = 2$

$\sin^n\theta + \csc^n\theta = 2$ Ans.

27. $\sin^6 x + \cos^6 x + 3\sin^2 x\cos^2 x = ?$

$(\sin^2 x)^3 + (\cos^2 x)^3 + 3\sin^2 x.\cos^2 x$

$(\sin^2 x + \cos^2 x)^3 - 3\sin^2 x\cos^2 x(\sin^2 x + \cos^2 x) + 3\sin^2 x\cos^2 x$

$\sin^2 x + \cos^2 x = 1$

$1^3 - 3\sin^2 x - \cos^2 x + 3\sin^2 x\cos^2 x$

$= 1$ Ans.

28. $x = a\cos^2\sin\theta$, $y = a\sin^2\theta\cos\theta$ $(x^2+y^2)^3 = ?$

$x^2 = a^2\cos^4\theta\sin^2\theta, y^2 = a^2\sin^4\theta\cos^2\theta$

$(x^2 + y^2) = a^2\cos^4\theta\sin^2\theta + a^2\sin^4\theta\cos^2\theta$

$(x^2 + y^2) = a^2\cos^2\theta\sin^2\theta(\sin^2\theta + \cos^2\theta)$

$(x^2 + y^2) = a^2\cos^2\theta\sin^2\theta$

$(x^2 + y^2)^3 = a6\cos^6\theta\sin^6\theta$

$(x^2 + y^2)^3 = a^2(a^2\cos^4\theta\sin^2\theta.\ a^2\sin^4\theta\cos^2\theta)$

$(x^2 + y^2)^3 = a^2x^2y^2$

29. $3\cos\theta + 4\sin\theta + 10$

$\min = -\sqrt{a^2+b^2} + c$

$-\sqrt{9+16} + 10$

` $-5 + 10$

$= 5$ min value.

30. $\sin 6^0 \sin 42^0 \sin 66^0 \sin 78$

$= \frac{1}{4}[(\cos 60^0 - \cos 72^0)(\cos 36 - \cos 120)]$

$= \frac{1}{4}\left[\left(\frac{1}{2} - \frac{\sqrt{5}-1}{4}\right)\left(\frac{\sqrt{5}+1}{4}\ \frac{1}{2}\right)\right]$

For $\frac{x^2}{9} + \frac{y^2}{4} = 1$

$\frac{1}{9} + \frac{2^2}{4} - 1 > 0$

And $\frac{2^2}{9} + \frac{1}{4} - 1 < 0$

=P lies outside E and Q …. Inside E

Ans. (d)

31. (c)

20^{th} term = 30

30^{th} term = 20

$a_{20} = a + 19d = 30$ …………(1)

$a_{30} = a + 29d = 20$…………(2)

subtract (i) from (2)

$(a+29d) - (a+19d) = 20-30$

$10d = -10$

$d = -1$

so $a + 19(-1) = 30$

$a = 30 + 19$

$= 49$

So 10^{th} term $= a + 9d$

$= 49 + 9(-1)$

32. $S_n = 2n(n-1)$

$n = 1$

$S_1 = 0 = a_1$

$a_1 = 0$

$n = 2$

$S_2 = 4(1) = 4 = a_1 + a_2$

$a_2 = 4$

$n = 3$

$S_3 = 6 \times 2 = 12 = a_1 + a_2 + a_3$

$a_3 = 8$

$n = 3$

$S_4 = 8 \times 3 = 24 = a_1 + a_2 + a_3 + a_4$

$a_4 = 12$

$(a_1, a_2, a_3, a_4 \ldots\ldots a_n)$

(0,4,8,12,16……..n)

4(0,1,2,3,4……..n)

$(4)^2(0+1^2+2^2+3^2+4^2\ldots\ldots\ldots\ldots n^2)$

$\frac{4^2[n(n+1)(2n+1)]}{6}$

$\frac{8[n(n+1)(2n+1)]}{3}$ Answer.

33. Let sum of 1^{st} m terms = 5m

& sum of 1^{st} n terms = Sn

So $\frac{5m}{5n} = \frac{m^2}{n^2}$

$= \frac{\frac{m}{2}[2a+(m-1)d]}{\frac{n}{2}[2a+(n-1d)} = \frac{m^2}{n^2}$

$\frac{2a+(m-1)d}{2a+(n-1)d} = \frac{m}{n}$

$n[2a + (m-1)d] = m[2a + (n-1)d]$

2an+mnd-nd+2am+mnd-nd

md-nd=2am-2an

d=2a

So Ratioof m^{th} and n^{th} term

$\frac{am}{an} = \frac{a+(m-1)d}{a+(n-1)d}$

$= \frac{a+(m-1)2a}{a+(n-1)2a}$

$= \frac{a(1+2m-2)}{a(1+2n-2)}$

$= \frac{2m-1}{2n-1}$

Ans. (c)

34. $9^{1/3} \times 9^{1/9} \times 9^{1/27} \times \ldots\ldots\ldots\ldots \infty$

$9\left(\frac{1}{3}+\frac{1}{9}+\frac{1}{27}+\cdots\ldots\ldots\infty\right)=9^{\frac{a}{(1-r)}}$

$a = 1/3 \quad r = 1/3$

$\frac{(1/3)}{9\left(1-\frac{1}{3}\right)} = 9^{1/2} = 3$

Ans. (c)

35. α, β roots of equation

$x^2 + px + p^2 + q = 0$

equation formed by 2 roots

$x^2 - (\alpha+\beta)x + \alpha\beta = 0$

So $\alpha+\beta = -P$

$\alpha\beta = P^2 + q$

So $(\alpha^2+\beta^2) + \alpha\beta$

$(\alpha^2+\beta^2) + \alpha\beta$

$\alpha^2+\beta^2 - 2\alpha\beta + \alpha\beta$

$(\alpha+\beta)^2 - \alpha\beta$

$(-P)^2 - (P^2 + q)$

$P^2 - P^2 - q$

$= -q$

37. $x^2 - bx + c = 0, \quad b^2 - 4c = ?$

root are consecutive number

$P + (P+1) = b \quad = \quad 2P + 1 = b$

$P(P+1) = C \quad = \quad P^2 + P = C$

$(2P+1)^2 - 4(P^2+P)$

$4P^2 + 4P + 1 - 4P^2 - 4P$

$= 1$

38. $(x-1)^2 + (x-2)^2 + (x-3) = 0$

$x^2 + 1 - 2x + x^2 + 4 - 4x + x^2 + 9 - 6x = 0$

$3x^2 - 12x + 14 = 0$

$D = b^2 - 4ac$

$= (-12)^2 - 4(3)(14)$

$= 144 - 168$

$= -24 < 0$

$\because D < 0$ No real roots of the equation.

Ans. (a)

39. $(b-c)x^2 + (c-a)x + (a-b) = 0$

General equation

$Ax^2 + Bx + C = 0$

$D = B^2 - 4AC = 0$

So $(c-a)^2 = 4(a-b)(b-c)$

$c^2+a^2-2ac = 4\{ab-ac-b^2+bc\}$

$c^2+a^2-2ac = 4ab - 4ac - 4b^2 + 4bc$

$a^2 + 4b^2 + c^2 + 2ac - 4ab - 4bc = 0$

$(a-2b-c)^2 = 0$

$a - 2b + c = 0$

$a + c = 2b$

$b = \frac{a+c}{2}$

So a, b & c are in AP.

Ans. (c)

40. $x^2 + 2x + 3\lambda = 0$

$2x^2 + 3x + 5\lambda = 0$

Condition for common root

$\frac{x^2}{10\lambda-9\lambda} = \frac{x}{5\lambda-6\lambda} = \frac{1}{3-4}$

$(6\lambda - 5\lambda)^2 = (3-4)(10\lambda - 9\lambda)$

$\lambda = -1$

Ans. (b) -1

41. ${}^nP_r = {}^nP_{r+1}$

$\frac{n!}{r!(n-r)!} = \frac{ni}{(r-1)!(n-r+1)!}$

$\frac{r}{n-r+1} = 1$

$n = 2r-1$

from equation (1)

$n = r + 1$

$= r + 1 = 2r - 1$

$r = 2$

$n = r + 1 = 3$

So $r = 2, \ n = 3$

Ans. (b) (3, 2)

42. Total no. of ways

$= \frac{6!}{2!3!} = 60$

No. of words in which 2N's come together is

$\frac{5!}{3!} = 20$

So Required no is $60 - 20 = 40$

Ans. (a)

43. $(1-x)^n = C_0 - C_1 x + C_2 x^2 - C_3 x^3 + n$ terms

So $x(1-x)^n = C_0x - C_1x^2 + C_2x^3 - C_2x^4 + \ldots\ldots$

$\int_0^1 x(1-x)^n\, dx$

$= \int_0^1 (C\, o\, x - c_1x^2 + c_2x^3 - c_3x^3 + \cdots.)\, dx$

$= -\int_0^1 (1-t)^n \ t\, dt$

$= \left[\left(C_0\frac{x^2}{2} - C_1\frac{x^3}{3} + C_2\frac{x^4}{x_4} - C_3\frac{x^5}{5} + \cdots.\right]_0^1\right.$

Where $r = 1 - x$

$-\left[\left(\frac{r^{n+1}}{n+1} - \frac{r^{1+2}}{n+2}\right]_1^0\right.$

$= \frac{C_0}{2} - \frac{C_1}{3} + \frac{C_2}{4} - \frac{C_3}{5} + \cdots.. \frac{1}{n+2} - \frac{1}{n+2}$

$= \frac{1}{(n+1)(n+2)}$

Ans. (d)

44. $\begin{bmatrix} 1 & w & w^2 \\ 1 & w^2 & 1 \\ w & 1 & w \end{bmatrix}$ $w^3 = 1$

$1 + w + w^2 - (w^2 + 1 + w^2) = 0$

45. $A = \begin{bmatrix} x & 2 \\ 2 & x \end{bmatrix}$

$A^2 = A.A.$

$A^2 = \begin{bmatrix} x.2 + 2.2 & x.2 + 2.x \\ 2.x + x.2 & 2.2 + x.x \end{bmatrix}$

$\begin{bmatrix} x^2 + 4 & 2x + 2x \\ 2x + x.2 & 4 + x^2 \end{bmatrix}$

$\begin{bmatrix} x^2 + 4 & 4x \\ 4x & x^2 + 4 \end{bmatrix}$

$|A^2| = (x^2+4)^2 - (4x)^2$

$= (x^2 + 4)^2 - 16x^2$

$= x^4 + 8x^2 + 16 - 16x^2$

$= x^4 - 8x^2 + 16$

Given $|A|^2 = 0$

$x^4 - 8x^2 + 16 = 0$

let $y = x^2$

$y^2 - 8y + 16 = 0$

$y = \frac{8 \pm \sqrt{64-64}}{2}$

$y = \frac{8}{2}$

$y = 4$

So $x = \pm 2$

Ans. (a) ± 2

46. $\vec{A} = i - \hat{j} + \hat{k}$

$\vec{C} = -\hat{\imath} - \hat{k}$

$\bar{A} \times \bar{B} = \bar{C}$

Let $\bar{B} = b_1\hat{\imath} + b_2\hat{\jmath} + b_3\hat{k}$

$\bar{A}.\bar{B} = 1$

$b_1 - b_2 + b_3 = 1$ ....(1)

$\bar{A} \times \bar{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 1 & -1 & 1 \\ b_1 & b_2 & b_3 \end{vmatrix}$

$-\hat{\imath} - \hat{\jmath} = \hat{\imath}\ (-b_3 - b_2) - \hat{\jmath}\ (b_3 - b_1) + k\ (b_2 + b_1)$

$-b_3 - b_2 = -1$

$b_3 + b_2 = 1$ .....(2)

$b_3 - b_1 = 1$ ....(3)

$b_2 + b_1 = 0$

$b_2 = -b_1$

$b_3 = 1 + b_1$

Put the value b_2 and b_3 equation (1)

$b_1 + b_1 + 1 + b_1 = 1$

$3b_1 = 0$

$b_1 = 0$

$b_3 = 1$

$b_2 = 0$

$\bar{B} = \hat{k}$ Answer.

47. P(0, K) ; A(–5, 4) ; B(3, –2)

PA = PB

$PA^2 = PB^2$

$25 + (k-4)^2 = 3^2 + (k+2)^2$

$25 + 16 + k^2 - 8k = 9 + 4 + k^2 + 4k$

$28 = 12k$

$K = \frac{28}{12} = \frac{7}{3}$

$P\left(0, \frac{7}{3}\right)$

Ans. (d)

48. $A = (x_1 y_1)$

$B = (x_2, y_2)$

$C = (x_3. y_3)$

Area $= \frac{1}{2}[x_1(y_3 - y_2) + x_2 (y_1 - y_3) + x_3 (y_2 - y_1)]$

So Area $= \frac{1}{2}[a(a + b - (c +a))+ b (b + c - (a+b))+C (C+a - (b+c))]$

Area $= \frac{1}{2}[ab - ac + bc - ba + ca - cb]$

Area = 0

Ans.

49. Let E be the event of getting the sum of 5 E = (1, 4), (4,1), (2,3), (3,2)

So n (E) = 4

Also n (s) = 6 x 6 = 36

Probability of event E

$P(E) = \frac{n(E)}{n(S)} = \frac{4}{36} = \frac{1}{9}$

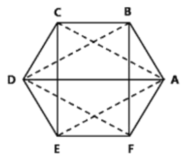
Ans. (c)

50. Let T represent the equilateral triangle n(T) = No. of equilateral triangle with six vertices ABCDEF of regular hexagon are only 2.

Namely Triangle ACE, Triangle BDF

Ans. n(S) = Total no. of Δ with b vertices are $^6C_3 = 20$

$P(T) = \frac{n(T)}{n(S)} = \frac{2}{20} = \frac{1}{10}$



Ans. (b)

51. (a)
52. (a)
53.
54. (d)
55. for input OO (decimal 0) : $0^3 = 0$
For input 01 (decimal 1) : $1^3 = 1$
For input 10 (decimal 2): $2^3 = 8$
For input 11 (decimal 3) : $3^3 = 27$
Ans. (b)

56.	(c)	57.	(b)	58.	(c)	59.	(d)
60.	(a)	61.	(b)	62.	(a)	63.	(c)
64.	(d)	65.	(d)				

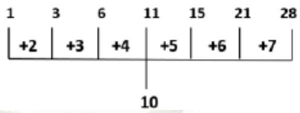
66. $(500)10 = (x)_5$
(1) 500 ÷ 5 = 100 remainder O
(2) 100 ÷ 5 = 20 remainder 0
(3) 20 ÷ 5 = 5 remainder 0
(4) 4 ÷ 5 = 0 remainder 4
So $500_{10} = 4000_5$
X in base 5 is 4000
Ans. (b)
67. $(780)_{10} = (1056)_x$
$(1056)\ x = 1.\ X^3 + 0.x^2 + 5.\ x + 6$
$= x^3 + 5\ x + 6$
$\because (1056)\ x = (780)_{10}$
So $x^3 + 5x + 6 = 780$
$x^3 + 5x – 774 = 0$
For x = 8
$8^3 + 5.8 –774$
512 + 40 – 774 = –222 (not zero)
For x = 9
$9^3 + 5.9 – 774$
= 729 + 45 – 774 = 0
x = 9

68. $(2x_1)_7 = (120)_{10}$
So $(2x_1)_7 = 2 \times 7^2 + x.\ 7^1 + 1.7^0$
= 98 + 7x + 1
$\because (2x_1)_7 = (120)_{10}$
So 98 + 7x + 1 = 120
7x = 21
X = 3
Ans. (c)

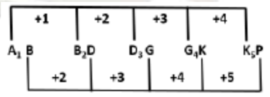
69. Given no. $(0110100)_2$
2's compliment
= 1's complement + 1
1's complement of 0110100
= 1001011
2's complement = 1001011 + 1
= 1001100
Ans. (a)

70. 10110010 no. is negative
MSB is 1 no. is negative
2's comp = 01001101 + 1
= 01001110
$(01001110)_2 = 78$
So negative value = –78
Ans. (b)

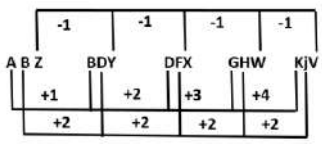
71. No. option.

72. $1^2 = 1$
$2^2 = 4$
$3^2 = 9$
$4^2 = 16$
$5^2 = 25$ (Not include in series)
$6^2 = 36$
$7^2 = 49$
Ans. (b)
73. 1,3,6,11,15,21,28


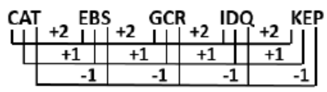
11 wrong number.

74. A_1B, B_2D, D_3G, G_4K.........?


Ans. K_5P

76. ABZ, BDY, DFX, GHW...........?


Ans. KJV

77. CAT, EBS, GCR, IDQ ....?


Ans. KEP.

78. 234 → 11
123 → ?

4 x 2 + 3 = 11
1 x 3 + 2 = 5 Ans.

79. 123456 → 615
214652 → ?
1 + 2 + 3 = 6, 4 + 5 + 6 = 15
2 + 1 + 4 = 7, 6 + 5 + 2 = 13
214652 = 713

80. 234 : 24 :: 235 : ?
2 x 3 x 4 = 24
2 x 3 x 5 = 30

81. 123 : 9 :: 321 : ?
(1 x 2) x 3 = 9
(3 + 2) x 1 = 5

82. JMI → 32
CAT → ?
J = 10 J + M + I = 32
M = 13 10 + 13 + 9 = 32
I = 9
C = 3 C + A + T = 24
A = 1 3 + 1 + 20 = 24
T = 20
CAT → 24 Ans.

83. BAG → 217
JMI → ?
B = 2 BAJ = 217
A = 1
G = 7
J = 10 JMI = 10139
M = 13
I = 9

84. CAT = 3
HE = 2 No. of Alphabet
DELHI = 5
SAD = 3
Ans. 3

85. 54 + 43 = 2 ⇒ 5 - 4 + 4 - 3 = 2
60 + 21 = 10 ⇒ 6 - 0 + 5 - 1 = 10
70 + 61 = 12 ⇒ 7 - 0 + 6 - 1 = 12
72 + 62 = ? ⇒ 7 - 2 + 6 - 2 = 9

86. 1, 3, 6, 11, 18, 29 .... ?
1 3 6 11 18 29 42
+2 +3 +5 +7 +11 +13
Ans. 42

87. 1, 8, 27, 64, 125, ...?
1^3 = 1
2^3 = 8
3^3 = 27
4^3 = 64
5^3 = 125
6^3 = 216
Ans. 216

88. 3, 7, 13, 21, 31, ... ?
3 7 13 21 31 43
+4 +6 +8 +10 +12
Ans. 43

89. 1, 2, 6, 42, ... ?
1 2 6 42 1806
1^2+1 2^2+2 6^2+6 42^2+42
Ans. 1806

91.	(b)	92.	(c)	93.	(a)	94.	(c)
95.	(c)	96.	(a)	97.	(c)	98.	(d)
99.	(a)	100.	(b)				